



(Year 2 / A2 Level)

Circular Motion Lecture Notes

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CHAPTER 12

Circular Motion

12.1 angular quantities

movement or rotation of an object along a circular path is called **circular motion**

to describe a circular motion, we can use *angular quantities*, which turn out to be more useful than linear displacement, linear velocity, etc.

12.1.1 angular displacement

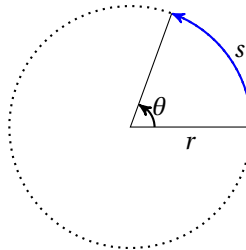
angular displacement is angle swiped out by object moving along circular

➤ unit: $[\theta] = \text{rad}$ (natural unit of measurement for angles)

conversion rule: $2\pi \text{ rad} = 360^\circ$

➤ if two radii form an angle of θ , then length of arc: $s = r\theta$

two radii subtending an arc of same length as radius form an angle of one **radian**



12.1.2 angular velocity

angular velocity describes how fast an object moves along a circular path

angular velocity is defined as angular displacement swiped out per unit time:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

➤ unit of: $[\omega] = \text{rad s}^{-1}$, also in radian measures

➤ angular velocity is a *vector* quantity

this vector points in a direction normal to the plane of circular motion

but in A-level course, we treat angular velocity as if it is a scalar

angular velocity and angular speed may be considered to be the same idea

➤ relation with linear velocity

in interval Δt , distance moved along arc $\Delta s = v\Delta t = r\Delta\theta \Rightarrow \omega = \frac{\Delta\theta}{\Delta t} = \frac{v}{r} \Rightarrow \boxed{v = \omega r}$

this relation between linear speed and angular speed holds at any instant

12.1.3 uniform circular motion

when studying linear motion, we started from motion with constant velocity v

consider the simplest possible circular motion $\rightarrow$ circular motion with constant ω

analogy with linear motion with constant v

uniform linear motion: $s = vt$

displacement $s \leftrightarrow \theta$, velocity $v \leftrightarrow \omega$

for uniform circular motion, one has: $\boxed{\theta = \omega t}$

➤ time taken for one complete revolution is called **period** T

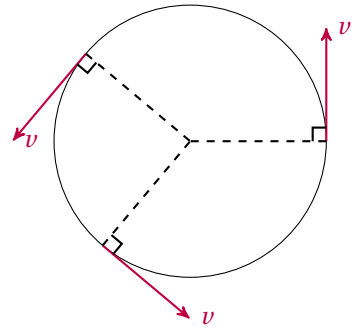
in one T , angle swiped is 2π , so $\boxed{\omega = \frac{2\pi}{T}}$

➤ uniform circular motion is still *accelerated* motion

speed is unchanged, but *velocity* is changing

direction of velocity always *tangential* to its path, so direction of velocity keeps changing

in general, any object moving along circular path is accelerating



Example 12.1 An object undergoes a uniform motion around a circular track of radius 2.5 m in 40 s, what is its angular speed and linear speed?

 $\omega = \frac{2\pi}{T} = \frac{2\pi}{40} \approx 0.157 \text{ rad s}^{-1} \quad v = \omega r = 0.157 \times 2.5 \approx 0.39 \text{ m s}^{-1} \quad \square$

Question 12.1 What is the angular velocity of the minute hand of a clock?

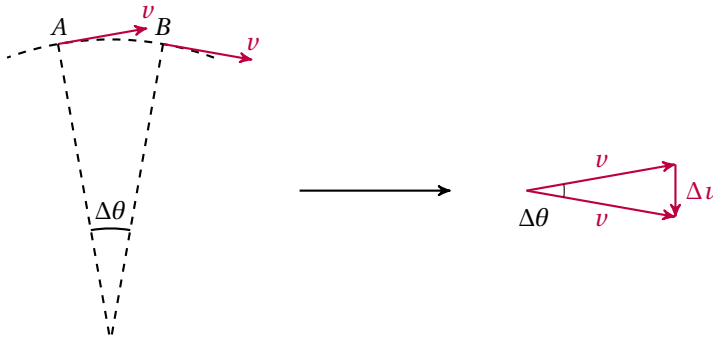
Question 12.2 A spacecraft moves around the earth in a circular orbit. The spacecraft has a speed of 7200 m s^{-1} at a height of 1300 km above the surface of the earth. Given that the radius of the earth is 6400 km. (a) What is the angular speed of this spacecraft? (b) What is its period?

12.1.4 centripetal acceleration

centripetal acceleration is the acceleration due to the change in direction of velocity vector, it points toward the centre of circular path

consider motion along a circular path from A to B with constant speed v

under small (infinitesimal) duration of time Δt ^[1]



change in velocity: $\Delta v = 2v \sin \frac{\Delta\theta}{2} \approx v\Delta\theta$ (as $\Delta\theta \rightarrow 0$, $\sin \Delta\theta \approx \Delta\theta$)

acceleration: $a = \frac{\Delta v}{\Delta t} \approx v \frac{\Delta\theta}{\Delta t} = v\omega$ (as $\omega = \frac{\Delta\theta}{\Delta t}$)

recall relation $v = \omega r$, we find centripetal acceleration: $a_c = \frac{v^2}{r} = \omega^2 r$

- direction of centripetal acceleration: always towards centre of circular path
- centripetal acceleration is only responsible for the change in *direction* of velocity

change in *magnitude* of velocity will give rise to *tangential acceleration*

this is related to *angular acceleration*^[2], which is beyond the syllabus

Question 12.3 A racing car makes a 180° turn in 2.0 s. Assume the path is a semi-circle with a radius of 30 m and the car maintains a constant speed during the turn. (a) What is the angular velocity of the car? (b) What is the centripetal acceleration?

^[1] A more rigorous derivation can be given by using differentiation techniques

^[2] Angular acceleration is analogous to linear acceleration a , defined as rate of change of angular velocity: $a = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ (*). Similar to $v = \omega r = \frac{ds}{dt}$, the relation $a = \alpha r = \frac{dv}{dt}$ also holds.

12.2 centripetal force

circular motion must involve change in velocity, so object is not in equilibrium

there must be a *net force* on an object performing circular motion

centripetal force (F_c) is the resultant force acting on an object moving along a circular path, and it is always directed towards centre of the circle

➤ centripetal force causes centripetal acceleration

using Newton's 2nd law: $F_c = m \frac{v^2}{r} = m\omega^2 r$

➤ F_c is not a new force by nature, it can have a variety of origins

F_c is a resultant of forces you learned before (weight, tension, contact force, friction, etc.)

➤ F_c acts at right angle to direction of velocity

or equivalently, if $F_{\text{net}} \perp v$ and F_{net} is of constant magnitude

then this net force provides centripetal force for circular motion

➤ effect of F_c : change *direction* of motion, or maintain circular orbits

to change *magnitude* of velocity, there requires a *tangential* component for the net force

again the idea of tangential force is beyond the syllabus

Example 12.2 A rock is able to orbit around the earth near the earth's surface. Let's ignore air resistance for this question, so the rock is acted by weight only. Given that radius of the earth $R = 6400$ km. (a) What is the orbital speed of the rock? (b) What is the orbital period?

✍ weight of object provides centripetal force: $mg = \frac{mv^2}{R}$

orbital speed: $v = \sqrt{gR} = \sqrt{9.81 \times 6.4 \times 10^6} \approx 7.9 \times 10^3 \text{ m s}^{-1}$

period: $T = \frac{2\pi R}{v} = \frac{2\pi \times 6.4 \times 10^6}{7.9 \times 10^3} \approx 5.1 \times 10^3 \text{ s} \approx 85 \text{ min}$ □

Example 12.3 A turntable can rotate freely about a vertical axis through its centre. A small object is placed on the turntable at distance $d = 40$ cm from the centre. The turntable is then set to rotate, and the angular speed of rotation is slowly increased. The coefficient of friction between the object and the turntable is $\mu = 0.30$. If the object does not slide off the turntable, find the maximum number of revolutions per minute.

if object stays on turntable, friction provides the centripetal force required: $f = m\omega^2 d$

increasing ω requires greater friction to provide centripetal force

but maximum limiting friction possible is: $f_{\text{lim}} = \mu N = \mu mg$, therefore

$$f \leq f_{\text{lim}} \Rightarrow m\omega^2 d \leq \mu mg \Rightarrow \omega^2 \leq \frac{\mu g}{d} \Rightarrow \omega_{\text{max}} = \sqrt{\frac{0.30 \times 9.81}{0.40}} \approx 2.71 \text{ rad s}^{-1}$$

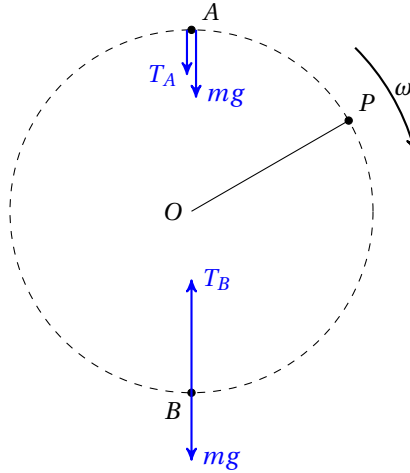
$$\text{period of revolution: } T_{\text{min}} = \frac{2\pi}{\omega_{\text{max}}} = \frac{2\pi}{2.71} \approx 2.32 \text{ s}$$

$$\text{number of revolutions in one minute: } n_{\text{max}} = \frac{t}{T_{\text{min}}} = \frac{60}{2.32} \approx 25.9$$

□

Example 12.4 Particle P of mass $m = 0.40 \text{ kg}$ is attached to one end of a light inextensible string of length $r = 0.80 \text{ m}$. The particle is whirled at a constant angular speed ω in a vertical plane.

(a) Given that the string never becomes slack, find the minimum value of ω . (b) Given instead that the string will break if the tension is greater than 20 N , find the maximum value of ω .



at top of circle (point A): $F_c = T_A + mg = m\omega^2 r \Rightarrow T_A = m\omega^2 r - mg$

at bottom of circle (point B): $F_c = T_B - mg = m\omega^2 r \Rightarrow T_B = m\omega^2 r + mg$

tension is minimum at A, but string being taut requires $T \geq 0$ at any point, so $T_A \geq 0$

$$m\omega^2 r - mg \geq 0 \Rightarrow \omega^2 \geq \frac{g}{r}$$

$$\omega_{\text{min}} = \sqrt{\frac{g}{r}} = \sqrt{\frac{9.81}{0.80}} \approx 3.5 \text{ rad s}^{-1}$$

tension is maximum at B, but string does not break requires $T \leq T_{\text{max}}$, so $T_B \leq T_{\text{max}}$

$$m\omega^2 r + mg \leq T_{\text{max}} \Rightarrow \omega^2 \leq \frac{T_{\text{max}}}{m} - \frac{g}{r}$$

$$\omega_{\text{max}} = \sqrt{\frac{T_{\text{max}}}{m} - \frac{g}{r}} = \sqrt{\frac{20}{0.40} - \frac{9.81}{0.80}} \approx 6.1 \text{ rad s}^{-1}$$

□

Example 12.5 A pendulum bob of mass 120 g moves at constant speed and traces out a circle of radius $r = 10$ cm in a horizontal plane. The string makes an angle $\theta = 25^\circ$ to the vertical. (a) What is the tension in the string? (b) At what speed is the bob moving?

 vertical component of tension T_y equals weight

$$T_y = mg \Rightarrow T \cos \theta = mg$$

$$T = \frac{mg}{\cos \theta} = \frac{0.12 \times 9.81}{\cos 25^\circ} \approx 1.3 \text{ N}$$

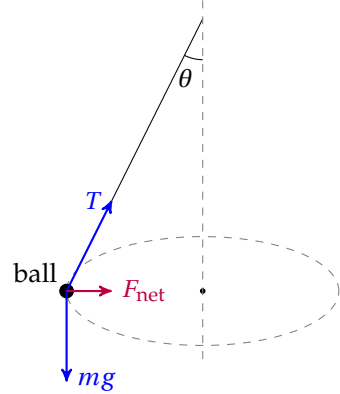
net force equals horizontal component of tension T_x

so component T_x provides centripetal force

$$F_c = T_x \Rightarrow T \sin \theta = \frac{mv^2}{r}$$

by eliminating T and m , one can find

$$v^2 = \frac{r \tan \theta}{g} = \frac{0.10 \times \tan 25^\circ}{9.81} \Rightarrow v \approx 0.069 \text{ m s}^{-1} \quad \square$$



Example 12.6 A small ball of mass m is attached to an inextensible string of length l . The ball is held with the string taut and horizontal and is then released from rest.

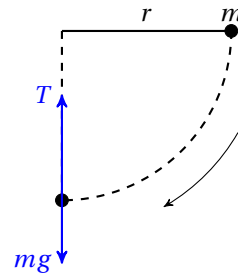
When the ball reaches lowest point, find its speed and the tension in the string in terms of m and l .

 energy conservation: G.P.E. loss = K.E. gain

$$mgr = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gr}$$

at lowest point: $F_c = T - mg = m \frac{v^2}{r}$

$$T = mg + m \frac{v^2}{r} = mg + m \frac{2gr}{r} = 3mg \quad \square$$



Question 12.4 Suggest what provides centripetal force in the following cases. (a) An athlete running on a curved track. (b) An aeroplane banking at a constant altitude. (c) A satellite moving around the earth.

Question 12.5 A turntable that can rotate freely in a horizontal plane is covered by dry mud. When the angular speed of rotation is gradually increased, state and explain whether the mud near edge of the plate or near the mud will first leave the plate?

Question 12.6 A bucket of water is swung at a constant speed and the motion describes a circle of radius $r = 1.0$ m in the vertical plane. If the water does not pour down from the bucket even when it is at the highest position, how fast do you need to swing the bucket?

Question 12.7 This question is about the design of a roller-coaster. We consider a slider that starts from rest from a point P and slides along a frictionless circular track as sketched below. P is at the same height as the top of the track T . (a) Show that the slider cannot get to T . (b) As a designer for a roller-coaster, you have to make sure the slider can reach point T and continue to slide along the track, what is the minimum height for the point of release?

